Listing 4.3 An example ad-hoc query written using SQL against data in a SQLite database with the schema shown in figure 4.6.

```
import sqlite3
import pandas as pd

conn = sqlite3.connect('maintenance.db') #1

query = '''
    SELECT WT.DESCRIPTION, WT.COMPLETED_DATE, WT.COMPLETED_BY
    FROM WORK_ORDER_TASK WT, WORK_ORDER W, ASSET A
    WHERE WT.STATUS = 'COMPLETED'
    AND WT.COMPLETED_BY = 'John Smith'
    AND WT.COMPLETED_DATE <= W.SCHEDULED_DATE
    AND WT.WORK_ORDER_ID = W.WORK_ORDER_ID
    AND W.ASSET_ID = A.ASSET_ID
    AND A.asset_id = 'F1'
'''

print("Tasks completed by John Smith on/before scheduled date for asset F1:")
results = pd.read_sql_query(query, conn) #2

print(results.to_string(index=False))

conn.close()
```

#1 SQLite stores its data in a file on the local filesystem.
#2 Pandas takes SQL query string and a connection to the database, and loads the results of the query to a dataframe.

Many software applications that model physical objects and systems use relational data models due to the integrity and consistency such a model enforces on the data as well as the flexibility in querying them via a standardized query language. You can use a relational model to represent data that benefits from ACID compliance and transactional integrity. If your digital twin integrates with other systems such as ERP, finance, or supply chain systems, many of these store their data in a relational model, and you will often find yourself querying this type of data.

4.3.2 Columnar structures

Columnar storage organizes data by measurement type (column) rather than by individual records (rows), which contrasts with traditional row-based databases that store complete records together.

Columnar storage offers several key advantages for analytical workloads and data processing. By storing data column-wise, it enables superior compression ratios since similar data types are grouped together, allowing compression algorithms to work more effectively. This structure dramatically improves query performance for analytical operations that typically access only a subset of columns—instead of reading entire rows, the system can retrieve just the specific columns needed for analysis. Columnar storage also